

intended to be reproduced. Moreover, because the sound waves would travel different paths through the listening space, the sound waves of an unmounted speaker would reach the listener at slightly different times, introducing annoying echo and reverberation.

Hence loudspeakers are not used without enclosures. The geometry of an enclosure also plays a role in managing vibration induced by the driver frame and moving airmass within the enclosure.

Enclosures can have adverse effect, unless these are well-designed, like panel resonance, diffraction from cabinet edges, and standing wave energy from internal reflection. These problems can be alleviated by choosing proper geometry of the enclosure such as avoiding sharp corners on the front of the enclosure. Curved speaker enclosures, which taper exponentially and are in the shape of a logarithmic spiral like nautilus mollusc, can address most of these problems.

Sometimes, the differences in the phase response at frequencies shared by different drivers can be solved by adjusting the vertical location of smaller drivers, so that the wavefront of all the drivers is coherent at or around the crossover frequencies in the speaker's normal sound field.

The box is typically made of wood, wood composite, sand mixed with resin, and more recently reinforced plastics. For damping the standing waves, fibreglass, felt made of jute, bonded acetate fibre, or wool can be used.

Ideally, a loudspeaker cone should respond equally to excitations throughout the frequency range so as not to favour one more than any other. However, the cone does have a set of natural frequencies, so when excited at these frequencies it will resonate, that is, respond more in that natural frequency than others. One way to address this problem is to make the speaker cone from a material that absorbs energy, like Kevlar, which is quite costly but helps in spreading the effect of resonance so that it is not so noticeable.

Another procedure is to divide the frequency range into, say, three or four parts and electronically allocate—preferably through active crossovers—each part of the frequency range to a sepa-

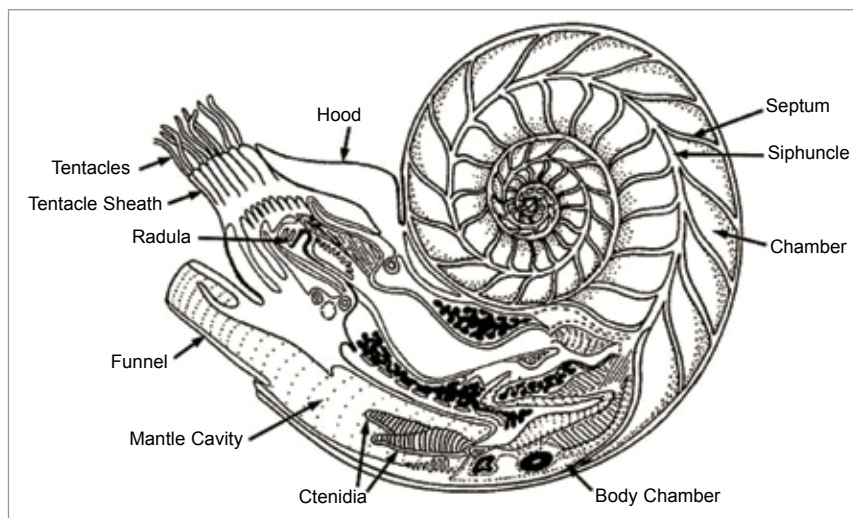


Fig. 2: A cross-section of nautilus

rate cone through a separate amplifier. By following this route, each speaker cone can be designed to cater for a limited range, and its shape and material can be so chosen that its natural frequencies do not spoil the sound.

Active crossover functions in a different mode than a passive one. Whereas passive networks split an amplified signal coming from one power amplifier so that it can be sent to multiple drivers, an active crossover splits up audio signal prior to amplification, enabling it to be sent to two or more power amplifiers, each of which is connected to a separate driver type. There is less likelihood of clipping, consistent crossover behaviour, and there is less loading on each amplifier, if active crossover is used. We can thus fine tune the whole system. Nautilus shaped speakers are generally provided with active crossovers to enhance their sound quality.

Transmission line

Nautilus shaped speakers are a variant of transmission line (TL) loudspeakers. A TL speaker has an enclosure design that uses the geometry of an acoustic transmission line within the cabinet. An acoustic transmission line uses a long duct for guiding sound waves, with minimum loss of energy, by restricting the transmission of energy to one direction without physical constraint. In this design, sound comes from speaker's back instead of reverberating as we see in simple box-type enclosure. It is directed into a long

folded or logarithmic spiral damped pathway within the speaker enclosure, which allows far greater control over the reproduction of sound and use of speaker energy, thereby improving the quality of resulting sound.

The pathway may vary in size and taper and may be open ended or close ended in a nautilus shaped speaker. Closed designs have negligible acoustic output from the enclosure, except from the driver. Such a design ensures that undesired resonances, reflections, and energies, which would otherwise cause undesirable auditory effects, are selectively damped because of the duct.

As a rule, a TL speaker tends to have exceptionally hi-fi low frequency response, far below subwoofer, reaching into infrasonic range as low as 17Hz without the need of a separate subwoofer driver. It creates a more spacious soundstage. The speaker's tapered cross-section avoids parallel internal surfaces that encourage standing waves. The enclosure behaves like an infinite baffle, potentially absorbing most of the speaker unit's rear energies.

TL speakers employ tube-like resonant cavity, with the length set between $1/6$ and $1/2$ the wavelength of the fundamental resonant frequency of the loudspeaker driver being used (generally $1/4$ for nautilus shape). The cross-section is typically tapered down at the terminus to about $1/4$ of the starting area. To match the sonic purity of tweeter and mid-range speaker, the bass needs a waveguide pipe of about